

NEWMAN, Australia

WMO: 943170

Lat: 23.4169S

Lon: 119.7989E

Elev: 1721

StdP: 13.80

Time Zone: 8.00 (E08)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.1	39.8	9.1	9.3	78.1	13.4	11.6	72.1	18.5	62.8	16.7	64.4	3.6	230	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	25.0	107.8	65.7	105.7	65.5	103.5	65.5	75.7	84.6	74.6	84.3	73.7	84.0	9.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	134.5	78.7	72.5	128.2	78.5	71.0	121.9	78.9	40.7	84.5	39.6	84.1	38.6	83.5	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	17.0	15.6	DB	33.0	110.7	2.7	2.2	31.1	112.3	29.5	113.6	28.0	114.9	26.1	116.5	(4)
(5)			WB	28.7	77.3	2.9	2.0	26.6	78.7	24.9	79.9	23.3	81.0	21.2	82.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	75.3	88.8	86.3	83.2	77.0	66.7	59.3	58.7	62.3	70.1	79.4	84.0	88.5	(6)
(7)		DBStd	12.27	6.09	5.78	5.98	5.35	5.93	5.37	4.94	5.19	5.70	5.58	5.11	4.75	(7)
(8)		HDD50	2	0	0	0	0	0	1	1	0	0	0	0	0	(8)
(9)		HDD65	582	0	0	0	2	52	186	204	117	20	1	0	0	(9)
(10)		CDD50	9232	1203	1016	1029	809	516	279	269	380	603	913	1021	1194	(10)
(11)		CDD65	4336	738	596	564	360	103	14	8	32	173	448	571	729	(11)
(12)		CDH74	62541	11094	8401	7505	4228	1287	205	274	844	2680	6276	8422	11325	(12)
(13)		CDH80	37056	7262	5148	4329	2117	423	21	27	211	1224	3624	5222	7448	(13)
(14)	Wind	WSAvg	7.6	8.7	8.4	8.0	6.7	6.4	6.4	6.2	6.9	7.8	8.3	8.7	8.9	(14)
(15)	Precipitation	PrecAvg	10.6	1.8	2.4	1.1	0.6	0.9	0.6	0.5	0.4	0.1	0.2	0.4	1.1	(15)
(16)		PrecMax	18.5	7.4	9.3	3.1	1.8	4.3	3.1	2.7	3.1	0.9	1.4	2.0	2.8	(16)
(17)		PrecMin	6.0	0.3	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	3.9	1.6	2.3	0.9	0.7	1.2	0.9	0.8	0.7	0.3	0.4	0.5	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	110.1	109.2	105.2	99.9	91.3	83.0	83.4	88.4	95.8	103.6	105.4	110.0	(19)
(20)			MCWB	66.0	64.8	66.4	63.9	60.3	59.2	58.4	58.4	59.3	62.3	63.0	65.8	(20)
(21)		2%	DB	107.9	106.5	101.9	96.1	88.2	79.0	79.8	84.9	92.9	100.1	103.2	107.0	(21)
(22)			MCWB	66.0	65.8	65.6	63.3	59.9	56.9	56.7	56.6	58.6	61.4	63.0	65.3	(22)
(23)		5%	DB	105.8	103.3	99.6	93.3	84.9	76.2	77.2	82.3	90.5	97.9	101.4	104.8	(23)
(24)			MCWB	66.4	66.7	65.9	63.4	59.1	55.8	55.0	55.0	58.2	60.8	62.9	64.9	(24)
(25)		10%	DB	103.0	100.0	96.7	90.7	81.5	73.3	74.1	79.1	87.6	95.4	99.1	102.4	(25)
(26)			MCWB	67.0	67.7	66.3	62.8	58.1	54.3	53.5	54.2	57.2	60.4	62.7	65.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.4	77.5	76.2	73.1	68.3	65.0	63.5	61.8	63.8	68.3	70.5	73.9	(27)
(28)			MCD B	86.0	84.6	85.2	81.4	78.8	73.3	75.2	76.7	82.8	83.2	84.1	84.3	(28)
(29)		2%	WB	75.7	75.9	74.3	71.3	64.5	61.7	60.2	59.7	61.4	66.4	68.1	72.6	(29)
(30)			MCD B	84.6	84.7	83.5	79.8	77.1	71.9	72.1	77.8	84.3	83.3	85.8	85.3	(30)
(31)		5%	WB	74.7	74.7	72.9	69.6	62.6	59.6	57.8	57.7	60.0	64.7	66.6	71.5	(31)
(32)			MCD B	84.4	84.5	82.5	80.1	76.4	70.0	71.5	78.3	84.8	84.3	88.9	86.0	(32)
(33)		10%	WB	73.7	73.7	71.6	67.5	60.5	57.2	55.7	55.9	58.6	62.9	65.4	70.2	(33)
(34)			MCD B	84.6	84.0	82.5	79.6	76.9	68.7	69.6	76.1	83.9	87.8	89.8	87.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	22.7	21.7	22.0	24.8	26.3	26.7	28.7	31.0	31.1	28.9	27.1	25.0	(35)
(36)			MCD BR	26.5	27.0	26.1	27.7	29.7	29.0	33.2	35.2	35.0	31.1	28.6	28.0	(36)
(37)			MCWBR	7.6	7.0	7.4	9.0	11.7	13.1	15.3	15.6	13.6	10.8	9.3	9.1	(37)
(38)		5% WB	MCD BR	20.2	18.7	18.4	19.9	22.5	21.3	26.2	29.9	30.6	25.6	26.0	22.3	(38)
(39)			MCWBR	5.6	5.6	6.2	7.2	10.1	10.6	13.1	14.1	12.7	8.9	8.2	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.364	0.359	0.355	0.339	0.311	0.296	0.301	0.304	0.324	0.349	0.352	0.353	(40)	
(41)		taud	2.550	2.582	2.556	2.559	2.557	2.585	2.548	2.507	2.456	2.429	2.458	2.530	(41)	
(42)		Ebn at Noon	311	308	301	292	288	287	289	299	305	307	312	314	(42)	
(43)		Edh at Noon	35	33	32	30	27	25	27	31	35	38	38	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2354	2226	1973	1714	1464	1312	1407	1790	2148	2366	2532	2527	(44)	
(45)		RadStd	171	165	127	96	112	108	110	65	74	115	100	140	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page